AGRIAI INTELLIGENCE SYSTEMS | CROP PROTECTION PLATFORM

System Installation & Configuration Guide

Prerequisites, Virtual Environments, and Local Setup

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Executive Summary: This document provides official specifications, standard operating procedures, and architectural blueprints for the AI-Powered Crop Pest Prediction & Agricultural Advisory System. The platform bridges machine learning multi-class diagnostics, agro-meteorological streaming data, and agronomic Integrated Pest Management (IPM) to protect farm yields sustainably.

1. Hardware & System Prerequisites

Resource	Minimum Requirement	Production Recommended
Operating System	Windows 10+, Ubuntu 20.04+, macOS 12+	Ubuntu 22.04 LTS x86_64
Processor	2 Cores (2.0 GHz)	4 Cores (3.0 GHz)
Memory (RAM)	4 GB	8 GB DDR4/DDR5
Storage	2 GB Free SSD Space	10 GB NVMe SSD
Python	Python 3.10.x - 3.13.x	Python 3.11.x 64-bit

2. Step-by-Step Installation Instructions

Step 1: Clone Repository

```
git clone https://github.com/Shubham-Prajapati-655/AI-Powered-Pest-Prediction.git
cd AI-Powered-Pest-Prediction
```

Step 2: Initialize Virtual Environment

```
# On Windows (PowerShell):
python -m venv venv
.\venv\Scripts\Activate.ps1

# On Linux/macOS:
python3 -m venv venv
source venv/bin/activate
```

Step 3: Install Pinned Dependencies

```
pip install --upgrade pip
pip install -r requirements.txt
```

Step 4: Launching Application Services

```
# Terminal 1: Streamlit Frontend & Multi-Page Dashboard
streamlit run app/app.py

# Terminal 2: FastAPI REST Microservice
uvicorn app.api:app --host 0.0.0.0 --port 8000
```

3. Automated Verification & Quality Assurance

Verify your local installation by executing the complete 91-test regression suite:

```
pytest tests/ -v
```

All tests across machine learning inference, database operations, weather fallbacks, and UI components should pass with a 100% success score.

